XRPL Validator — Daily Engineering Report · 2026-08-04

Period: 2026-08-04 00:00 to 24:00 US Eastern (2026-08-04 04:00 to 2026-08-05 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 1 commit merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
14:09 UTC	xrpl-validator	`9a877b7`	feat(loop): track rate.py, and stop counting hydration failures as divergences

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ledger): price an AMM bridge leg at the amount that actually flows** (`b18410f`, xrpl-validator) Four changes to the bridged crossing, all read off libxrpl's trace on #105912454 FE592890B233 (24 mutations against mainnet's 17).
- **feat(loop): track rate.py, and stop counting hydration failures as divergences** (`9a877b7`, xrpl-validator) rate.py is the native engine's real score — divergences per 1000 tx on UNSELECTED fresh ledgers, as opposed to what is left in a batch of scout-finds, which is the hard residue by construction and barely moves when a common root cause is fixed.
- **fix(ledger): a bridged leg's pool prices the strand only while two are active** (`edf469a`, xrpl-validator) #105930662 40FB322EC16C: 6 mutations against 13, all 7 missing ones Modified, nothing extra. Now 45/45 — the ledger is fully clean.
- **fix(ledger): the stream's dead tests apply to the taker's own offer too** (`37d5625`, xrpl-validator) #105949459 4A03010A4B1E: 4 mutations against 7. Now 178/178 — the ledger is fully clean.
- **feat(ledger): DX_HEADS/DX_RAWHEADS — per-leg book heads in the bridged walk** (`40cabb2`, xrpl-validator) Diagnostics only, both behind the existing `DX_BRIDGE` env var, nothing evaluated unless it is set. Added while chasing #105940336 CA2C624ED031 and kept because they answered two questions in one run each.

Independence program

No independence record for UTC day 2026-08-04; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Bonded Oracle Providers With Slashing** (oracles) Add sfBondAmount and sfBondAccount to the PriceOracle ledger object, funded at creation via a new OracleBond transaction that escrows XRP against the
- **Fixed-Term Amortizing Loans On Vaults** (lending) Add a new ltLOAN ledger object (owner-reserved, keylet::loan(vault, borrower, sequence)) carrying sfPrincipal, sfInterestRate, sfMaturityLedger, sfPay
- **Segregated Sub-Accounts With Proof-of-Reserves** (custody) Add an lsfSegregated flag and a SubAccountCreate transaction that derives a pseudo-account via keylet::subAccount(parent, index) (extending XLS-64), c

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-04 05:30	—	xrpl-validator	`b18410f`	fix(ledger): price an AMM bridge leg at the amount that actually flows
08-04 14:09	08-04 14:09	xrpl-validator	`9a877b7`	feat(loop): track rate.py, and stop counting hydration failures as divergences
08-04 15:25	—	xrpl-validator	`edf469a`	fix(ledger): a bridged leg's pool prices the strand only while two are active
08-04 21:46	—	xrpl-validator	`37d5625`	fix(ledger): the stream's dead tests apply to the taker's own offer too

Committed	Merged into main	Repo	Commit	Subject
08-04 22:37	—	xrpl-validator	`40cabb2`	feat(ledger): DX_HEADS/DX_RAWHEADS — per-leg book heads in the bridged walk

Times in UTC. A commit appears on the day it was committed to main or merged into it.